SENIOR SECONDARY 1 MATHEMATICS

A Comprehensive Guide Aligned with the NERDC Curriculum

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For use in Nigerian Senior Secondary Schools

Contents

Preface

This textbook has been written to provide Senior Secondary 1 students with a clear, thorough, and engaging treatment of mathematics as prescribed by the NERDC/WAEC curriculum.

Each chapter is structured to move from **definitions and concepts** through **worked examples** to **exercises**, giving the student a complete learning arc for every topic.

How to Use This Book

Work through every worked example with pen and paper before attempting the exercises. Mathematics is a practical subject — reading alone is never enough.

The Author

Course Outline (SS1 Curriculum)

First Term

1. Number Bases
2. Modular Arithmetic
3. Indices and Logarithms
4. Set Theory

Second Term

5. Algebraic Processes
6. Simple Equations and Variations
7. Logical Reasoning
8. Plane Geometry

Third Term

9. Trigonometry
10. Mensuration
11. Statistics and Probability

Part I

First Term

Chapter 1

Number Bases

Chapter introduction goes here.

1.1 Intro And Expansion

Definition: Term

Definition text here.

Example 1.1

Problem statement.
Solution:
Step-by-step solution here.

1.2 Conversion Of Bases

Definition: Term

Definition text here.

Example 1.2

Problem statement.
Solution:
Step-by-step solution here.

1.3 Basic Operations

Definition: Term

Definition text here.

Example 1.3

Problem statement.
Solution:
Step-by-step solution here.

Chapter Exercises

Exercise

1.
2.
3.

Chapter 2

Modular Arithmetic

Chapter introduction goes here.

2.1 Concept Of Modulo

Definition: Term

Definition text here.

Example 2.1

Problem statement.
Solution:
Step-by-step solution here.

2.2 Operations In Modulo

Definition: Term

Definition text here.

Example 2.2

Problem statement.
Solution:
Step-by-step solution here.

2.3 Applications

Definition: Term

Definition text here.

Example 2.3

Problem statement.
Solution:
Step-by-step solution here.

Chapter Exercises

Exercise

1.
2.
3.

Chapter 3

Indices and Logarithms

Chapter introduction goes here.

3.1 Laws Of Indices

Definition: Term

Definition text here.

Example 3.1

Problem statement.
Solution:
Step-by-step solution here.

3.2 Standard Form

Definition: Term

Definition text here.

Example 3.2

Problem statement.
Solution:
Step-by-step solution here.

3.3 Laws Of Logarithms

Definition: Term

Definition text here.

Example 3.3

Problem statement.
Solution:
Step-by-step solution here.

Chapter Exercises

Exercise

1.
2.
3.

Chapter 4

Set Theory

Chapter introduction goes here.

4.1 Set Notation And Types

Definition: Term

Definition text here.

Example 4.1

Problem statement.
Solution:
Step-by-step solution here.

4.2 Venn Diagrams

Definition: Term

Definition text here.

Example 4.2

Problem statement.
Solution:
Step-by-step solution here.

4.3 Union Intersection Complement

Definition: Term

Definition text here.

Example 4.3

Problem statement.
Solution:
Step-by-step solution here.

Chapter Exercises

Exercise

1.

2.

3.

Part II

Second Term

Chapter 5

Algebraic Processes

Chapter introduction goes here.

5.1 Factorisation

Definition: Term

Definition text here.

Example 5.1

Problem statement.
Solution:
Step-by-step solution here.

5.2 Algebraic Fractions

Definition: Term

Definition text here.

Example 5.2

Problem statement.
Solution:
Step-by-step solution here.

5.3 Substitution

Definition: Term

Definition text here.

Example 5.3

Problem statement.
Solution:
Step-by-step solution here.

Chapter Exercises

Exercise

1.
2.
3.

Chapter 6

Simple Equations and Variations

Chapter introduction goes here.

6.1 Linear Equations

Definition: Term

Definition text here.

Example 6.1

Problem statement.
Solution:
Step-by-step solution here.

6.2 Intro To Quadratics

Definition: Term

Definition text here.

Example 6.2

Problem statement.
Solution:
Step-by-step solution here.

6.3 Variation

Definition: Term

Definition text here.

Example 6.3

Problem statement.
Solution:
Step-by-step solution here.

Chapter Exercises

Exercise

1.
2.
3.

Chapter 7

Logical Reasoning

Chapter introduction goes here.

7.1 Simple Statements

Definition: Term

Definition text here.

Example 7.1

Problem statement.
Solution:
Step-by-step solution here.

7.2 Negation And Connectives

Definition: Term

Definition text here.

Example 7.2

Problem statement.
Solution:
Step-by-step solution here.

7.3 Conditional Statements

Definition: Term

Definition text here.

Example 7.3

Problem statement.
Solution:
Step-by-step solution here.

Chapter Exercises

Exercise

1.

2.

3.

Chapter 8

Plane Geometry

Chapter introduction goes here.

8.1 Angles At A Point

Definition: Term

Definition text here.

Example 8.1

Problem statement.
Solution:
Step-by-step solution here.

8.2 Parallel Lines

Definition: Term

Definition text here.

Example 8.2

Problem statement.
Solution:
Step-by-step solution here.

8.3 Triangles And Polygons

Definition: Term

Definition text here.

Example 8.3

Problem statement.
Solution:
Step-by-step solution here.

Chapter Exercises

Exercise

1.

2.

3.

Part III

Third Term

Chapter 9

Trigonometry

Chapter introduction goes here.

9.1 Trig Ratios

Definition: Term

Definition text here.

Example 9.1

Problem statement.
Solution:
Step-by-step solution here.

9.2 Solving Right Triangles

Definition: Term

Definition text here.

Example 9.2

Problem statement.
Solution:
Step-by-step solution here.

9.3 Elevation And Depression

Definition: Term

Definition text here.

Example 9.3

Problem statement.
Solution:
Step-by-step solution here.

Chapter Exercises

Exercise

1.

2.

3.

Chapter 10

Mensuration

Chapter introduction goes here.

10.1 Perimeter And Area

Definition: Term

Definition text here.

Example 10.1

Problem statement.
Solution:
Step-by-step solution here.

10.2 Surface Area

Definition: Term

Definition text here.

Example 10.2

Problem statement.
Solution:
Step-by-step solution here.

10.3 Volume

Definition: Term

Definition text here.

Example 10.3

Problem statement.
Solution:
Step-by-step solution here.

Chapter Exercises

Exercise

1.
2.
3.

Chapter 11

Statistics and Probability

Chapter introduction goes here.

11.1 Data Collection And Measures

Definition: Term

Definition text here.

Example 11.1

Problem statement.
Solution:
Step-by-step solution here.

11.2 Charts And Graphs

Definition: Term

Definition text here.

Example 11.2

Problem statement.
Solution:
Step-by-step solution here.

11.3 Probability

Definition: Term

Definition text here.

Example 11.3

Problem statement.
Solution:
Step-by-step solution here.

Chapter Exercises

Exercise

1.

2.

3.

Answers to Selected Exercises

Answers will be populated as chapters are written.

Glossary

Base The number of distinct digits used in a positional numeral system.

Congruence Two integers are congruent modulo n if their difference is divisible by n.

Index (Exponent) The power to which a base is raised; e.g. in a^n, n is the index.

Logarithm The inverse operation to exponentiation; $\log_b x = y \iff b^y = x$.

Set A well-defined collection of distinct objects called elements.

Variation A relationship between two quantities where one changes in proportion to the other.

Probability A measure of how likely an event is to occur, expressed as a value between 0 and 1.

Expand this glossary as new terms are introduced throughout the book.